

D(4th Sm.)-Computer Sc.-H/CC-10/CBCS

2025

COMPUTER SCIENCE — HONOURS

Paper : CC-10

(Microprocessor and its Application)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
 - (a) What is the function of HLDA?
 - (b) What is the function of Program Counter?
 - (c) Which interrupts in the microprocessor 8085 is maskable? What is the purpose?
 - (d) What is the purpose of ALE?
 - (e) What are the maximum address lines required if a memory of 16 K byte is connected to microprocessor 8085?
 - (f) How many flags are present in the flag register of the 8085 microprocessor? What is the significance of these flags in the functioning of the processor?
 - (g) What is the function of DAA? Give example.
 - (h) Explain the function of XCHG instruction.
2. (a) Draw the timing diagram of the instruction STA 8050H, assuming that the instruction is written across the memory location C000H, C001H and C002H and the operational code of STA is 32H. 8+2
 - (b) Explain the operation of PUSH PSW instruction.
3. (a) Describe the operations performed by the following 8085 microprocessor instructions :

RAR
 CMA
 DADrp.

 - (b) What are vectored interrupts with respect to 8085? Give example. 6+4

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4. (a) Explain the operation of the following assembly language program for microprocessor 8085. Show all the steps and the registers contents after execution of each instructions where memory location 8000H contains 87H data.

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LXI    H, 8000H
MOV     A, M
MOV     B, A
ADD     A
ADD     B
INX     H
MOV     M, A
HLT.

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- (b) Explain nesting of subroutines with suitable example. 6+2+2
- (c) What is the purpose of READY signal? 6+4
5. (a) Draw the internal architecture of microprocessor 8085 and explain in brief its various functional blocks.
- (b) Explain how ALE help in de-multiplexing of lower order address data bus $AD_0 - AD_7$ with suitable example. 6+4
6. (a) Explain foldback memory with respect to microprocessor 8085 with suitable example.
- (b) Write the functions of STAX and LDAX. 8+2
7. (a) What are the differences between 'Burst mode' and 'Cycle stealing' techniques of DMA data transfer scheme?
- (b) What are the functions of PC and SP? 5+5
8. (a) What is nested subroutine? Explain PUSH and POP operations?
- (b) Differentiate between I/O mapped I/O and memory mapped I/O with respect of 8085 microprocessor. (1+2+2)+5